

Your name: _____, Roll no.: _____

CSE 350/550: Network Security, Mid-semester exam: Feb 23, 2026, 90 minutes, 90 marks**Instructions:**

- The exam is in **two** parts: Part 1 has 5 questions that require short answers. Each question is for 6 marks, totaling 30 marks. Part 2 has 5 questions that require longer answers. Each question is of 12 marks, totaling 60 marks.
- This is a “closed-book” exam. All necessary information, including tables, etc. will be provided to you.
- Write your answers in the space provided. If necessary, use a separate sheet for rough work.
- Do not use unfair means. Action will be taken against you if you use unfair means. You may not use books, notes, computer, smart phone, etc. or share info with other students. You may only use a calculator.

Part 1: 5 questions that require short answers (6 marks each)

1. We give below the overall scheme for a “certification authority” (CA) to generate a “public-key certificate” $CERT_A$ for user A. The public-key is PU_A .

Marks
on Q1

$$CERT_A = [ID_A, PU_A, T_A, DUR_A, INFO_{CA}, ENC_{PR-CA}(\text{Hash}(ID_A, PU_A, T_A, DUR_A, INFO_{CA}))]$$

- A. How can an intruder use this certificate $CERT_A$ to generate a certificate with A’s fake public-key say, PU_X ?

Simply replace PU_A by PU_X . Nothing more. Verification of certificate will fail unless the signature (or the Hash value) remains the same when PU_A is replaced by PU_X .

3

- B. What is that failed assumption that allows the intruder to generate a certificate with A’s fake public-key?

The failed assumption is: the signature (as also the Hash value) will change if any of the components in $(ID_A, PU_A, T_A, DUR_A, INFO_{CA})$ is changed, or in particular if PU_A is replaced by PU_X .

3

2. This has to do with “mix-column” operation in an AES round, but limited to 16-bit plaintext/ciphertext, re-written as 2×2 state matrix of 4-bit nibbles. The 4-bit numbers belong to $GF(2^4)$. The add (+) and multiple (*) operations are defined in $GF(2^4)$, with the irreducible polynomial $m(x, 4) = x^4 + x + 1$.

Marks
on Q2

What is the resulting 2×2 matrix, post mix-column step? Just fill the missing entries in hexadecimal notation?

$$\begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix} * \begin{bmatrix} 6 & 4 \\ C & 0 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ P=7 & Q=3 \end{bmatrix}$$

$$P = 4 * 6 + 1 * C = 0100 * 0110 + 0001 * 1100$$

$$= x^2(x^2+x) + x^3+x^2 = x^4+x^2 \pmod{m(x)} = x^2+x+1$$

$$\text{Thus } P = 0111 = 7$$

$$Q = 4 * 4 + 1 * 0 = 0100 * 0100 + 0001 * 0000$$

$$= x^2(x^2) = x^4 \pmod{m(x)} = x+1$$

$$\text{Thus } Q = 0011 = 3$$

2 each for
detailed
calculations
of P and Q

1 each for
correct P
and Q

Roll no.: _____

Marks
on Q3

3. Consider DES, and let plaintext length $n=64$. Further, let $N = 2^{64}$. Assume we have t plaintext-ciphertext pairs $P_i, C_i = E(K, P_i)$, where we assume that the key K is a randomly chosen 56-bit string. Imagine that we wish to find K by exhaustive search. We could generate a 56-bit key K' and test whether $C_i = E(K', P_i)$ for $1, \dots, i, \dots, t$. If K' encrypts each P_i to its proper C_i , then we have some evidence that $K = K'$. However, it may be the case that the mappings $E(K, \cdot)$ and $E(K', \cdot)$ exactly agree on the t plaintext-ciphertext pairs P_i, C_i , but may or may not agree on other pairs. Such keys are called spurious keys.

Question: What is the probability that $E(K, \cdot)$ and $E(K', \cdot)$ agree on t or more plaintext-ciphertext pairs?

Before you jump to the final answer, give the result when $t=1$, then $t=2$. And then generalize it. If necessary, assume $t \ll N$.

- Prob. that $E(K, \cdot)$ and $E(K', \cdot)$ agree when $t=1$: **is $1/N$**
- Prob. that $E(K, \cdot)$ and $E(K', \cdot)$ agree when $t=2$: **is $(1/N)^2$**
- Prob. that $E(K, \cdot)$ and $E(K', \cdot)$ agree for any t : **is $(1/N)^t$**

2 each for
parts a, b, c

Marks
on Q4

4. User A wishes to send a 100 MB photo file to user B confidentially. Given that the photo file is large (of order 100 MB), user A has decided to use symmetric-key cryptosystem to encrypt the photo file. It is important to note that there is no session key (or a secret) currently established between A and B. Surely, A possesses the public-key of B, viz. PU_B , and B possesses its corresponding private-key, PR_B .

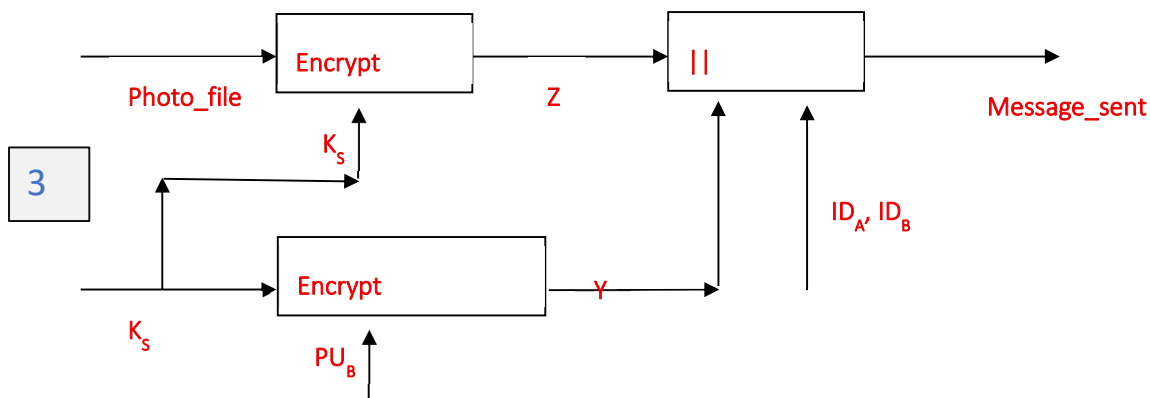
How can A communicate the photo file to B suitably encrypted using symmetric-key encryption.

- Explain briefly why this approach is computationally efficient.

Symmetric key encryption is more efficient compared to public-key encryption because symmetric-key encryption involves "shift", XOR, or table look-up operations, while public-key encryption involves multiplication of large numbers.

2

- Give a block diagram showing processing at sender's end.



- Specify the exact contents of the encrypted message sent, using the format given below. Do explain the various symbols/names used.

Message_sent = $\langle ID_A, ID_B, \dots, E(\dots, \text{Photo_file}) \rangle$

Message_sent = $\langle ID_A, ID_B, E(PU_B, K_s), E(K_s, \text{Photo_file}) \rangle$

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Marks
on Q5

5. Consider an RSA based public key cryptosystem, where the information available to an intruder is: the public key $\{e = 7, n = 187\}$, and ciphertext $C = 11$.
How can an intruder recover the plaintext M that was encrypted by user? Provide as many details as necessary. You may need to use this table to do your calculations.

	Quotient, $Q(i)$	Remainder, $R(i)$	$S(i)$	$T(i)$
		$R(i+1) = R(i-1) - Q(i+1)R(i)$	$S(i+1) = S(i-1) - Q(i+1)S(i)$	$T(i+1) = T(i-1) - Q(i+1)T(i)$
0		$\Phi = 160$	1	0
1		$e = 7$	0	1
2	22	6	1	-22
3	1	1	-1	$d = 23$
4	6	0		
5				
6, etc.				

Given $e = 7, n = 187, C = 11$

Step 1: Factor $n: 187 = 11 \times 17$

Step 2: Compute $\phi(n) = (11 - 1)(17 - 1) = 10 \times 16 = 160$

Step 3: Compute d using the above table: $d = 23$

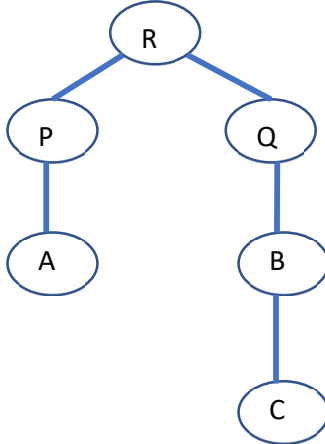
Step 4: Decrypt C :

$$\begin{aligned}
 M &= C^d \bmod n = 11^{23} \bmod 187 \\
 &= 11^{(16+4+2+1)} \bmod 187 \\
 &= 154 * 55 * 121 * 11 \bmod 187 \\
 &= 88
 \end{aligned}$$

1 marks each for
steps 1, 2;
2 marks each for
steps 3, 4

Part 2: 5 questions that require somewhat longer answers (12 marks each)Marks
on Q6

6. Consider the collection of entities A, B, C, P, Q and R that are either certification authorities or are users of certificates issued by such authorities. This is shown in figure below, where a parent node corresponds to the authority which originally signs the certificate of the child node. Since A does not possess the public-key of B, viz. PU_B , it is unable to verify the certificate of C, $CERT_{C-B}$, issued and signed by B. (Use notation: the public key of entity X is denoted as PU_X and certificate of entity X signed by Y is $CERT_{X-Y}$.)



- (a) Given that entity A has C's certificate signed by B, viz. $CERT_{C-B}$, what other information should entities A, B, C, P, Q and/or R possess before messages are exchanged between A, B, C, P, Q and R, the purpose being A verifying C's certificate signed by B, viz. $CERT_{C-B}$, and obtaining the public-key of C, PU_C ?

A: PU_P
 B: PU_Q
 C: PU_B
 P: PU_R
 Q: PU_R
 R: ----

Total of 3
marks for this

- (b) Now describe the steps or activities that entities A, B, C, P, Q and/or R undertake to ensure that A is able to obtain the public-key of C, viz. PU_C .

Fact: A has certificate of C signed by B, viz. $CERT_{C-B}$, but cannot verify it since it does not have PU_B

Steps undertaken:

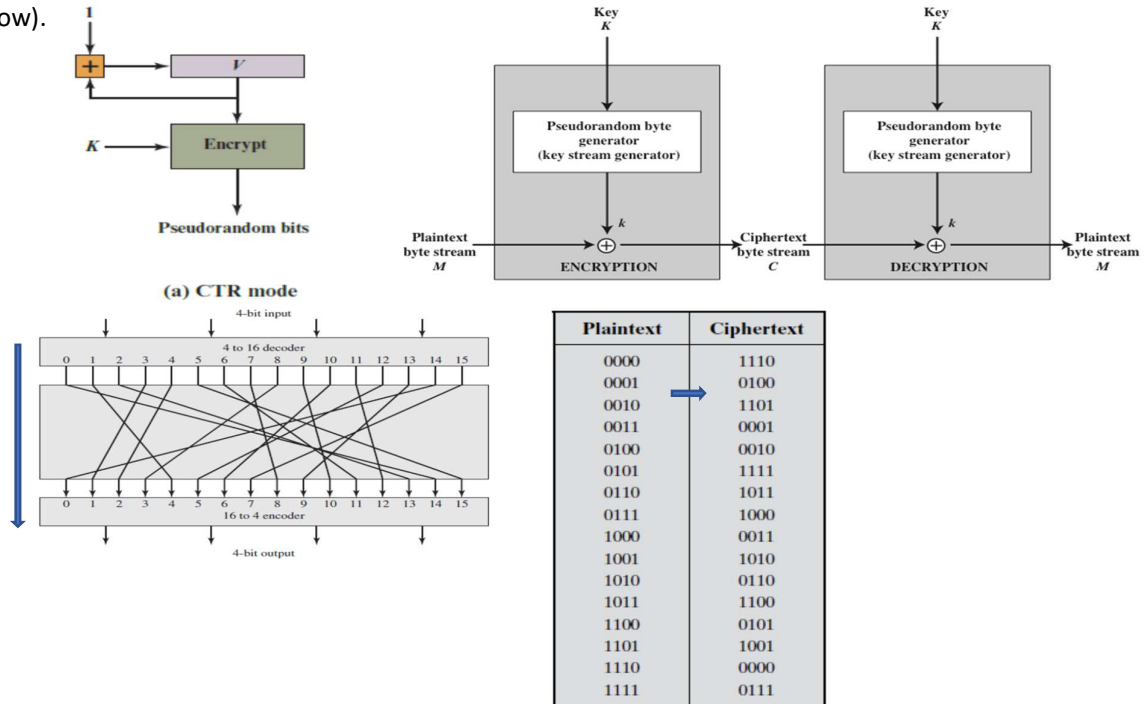
- A seeks from P B's certificate but one that is signed by P
- P seeks and obtains certificate from B. This happens to be signed by Q.
- P seeks and obtains certificate from Q. This happens to be signed by R.
- Using PU_R , P verifies $CERT_{Q-R}$ signed by R, obtains public key PU_Q , and adds PU_Q to its "directory"
- Using PU_Q , P verifies $CERT_{B-Q}$ signed by Q, obtains public key PU_B , and adds PU_B to its "directory"
- P creates a certificate $CERT_{B-P}$ containing public-key PU_B , signs it with its own PR_P , and delivers it to A
- Using PU_P A verifies C's certificate $CERT_{C-P}$ signed by P, and obtains PU_C
- Verifies C_B using PU_Q to obtain PU_B

Total of 9 marks
if the sequence is
correct, or 4
marks if it is
partly correct

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Marks
on Q7

7. Consider using a 'stream cipher', that encrypts a data stream of 4 bit 'nibbles', but using a 4-bit key-stream obtained from a pseudo-random generator (PRNG) based on 'Counter-mode encryption (CTR mode)' such as one described below. Here V , a 4-bit nibble, is initialized to 0, and increments by 1 (modulo 16) upon each tick. The encryption algorithm responsible for generating the stream of pseudorandom 4-bit nibbles is the "Mono-alphabetic substitution cipher", also described below (at the bottom). The stream cipher simply does an exclusive-OR to compute the ciphertext nibble given the plaintext and the random number (as shown below).



- What are the first SIX 4-bit nibbles generated by the CTR mode random-number generator, and corresponding to the first values of V . To answer, fill the table below.
- What is the periodicity of this PRNG? **16**
- For the given plaintext stream consisting of SIX nibbles, what is the ciphertext generated. To answer the question, fill the table below.
- If at any point an adversary is able to discover the k -th random number, then can he/she predict the following $k+1$ th random number or the $k+2$ th random number? **No. He/she cannot since the PRNG is a permutation of the 4 bits.**
- What is that information that the adversary should possess or compute which will enable him/her to predict the following $k+1$ th random number or the $k+2$ th random number? **He/she should know permutation used to scramble the 4-bit input.**

Serial no.	Plaintext	V	Pseudo random no.	Ciphertext
1	0011	0	1110	1101
2	1010	1	0100	1110
3	1110	2	1101	0011
4	0001	3	0001	0000
5	1010	4	0010	1000
6	0110	5	1111	1001

2 marks each for
correct columns in
table for V , random
numbers, and
ciphertext

Marks
on Q8

8. This is about exploring the possibility of breaking into an RSA based public-key cryptosystem. Here is one possibility. In a given public-key cryptosystem, it is known that $n = 391$. Two users, A and B, are assigned different pairs of public-private keys. The public keys are $PU_A = e_1 = 3$, while $PU_B = e_2 = 7$. Their private keys are not known publicly. On one occasion, and coincidentally, users A and B end up encrypting the same message M using their respective public-keys to obtain ciphertexts, $C_1 = 74$ and $C_2 = 279$. If an adversary were to get hold of values of C_1 and C_2 , how can he/she figure out M . Note the adversary has only the following information:

$n = 391$, $PU_A = e_1 = 3$, while $PU_B = e_2 = 7$, $C_1 = 74$ and $C_2 = 279$. **Hint: $e_2 = 2 * e_1 + 1$.**

DO NOT try to factor $n = 391$ into its factor. Since n is small, hacking into M is trivial.

Show the entire working, viz. what are C_1 and C_2 . And how is M related to C_1 and C_2 , given that $e_2 = 2 * e_1 + 1$.

Clearly $C_1 = M^{e_1} \bmod n = M^3 \bmod 391 = 74$

And $C_2 = M^{e_2} \bmod n = M^7 \bmod 391 = 279$

From above $C_2 = C_1^2 * M \bmod 391$, or $279 = 74^2 * M \bmod 391$

$\rightarrow M = 279 * (74^{-1})^2 \bmod 391$

Clearly since $74 * 37 \bmod 391 = 2738 \bmod 391 = 7 * 391 + 1 = 1$, $74^{-1} = 37$ (see table below)

$\rightarrow M = 279 * (74^{-1})^2 \bmod 391 = M = 279 * (37)^2 \bmod 391 = 381951 \bmod 391 = 335$. Final answer $M = 335$.

6 marks for recognizing the formula $M = 279 * (74^{-1})^2 \bmod 391$, and 6 marks for detailed calculations leading to $M=335$.

However, $335^3 \bmod 391 = 334$, not 74. Similarly $335^7 = 262$, not 279. (Check this out for yourself.) The problem is that C_1 was incorrectly specified. This should have been $C_1 = 147$, while n , e_1 , e_2 , C_2 are unchanged. The attack proceeds as follows:

Let $n=391$, $e_1=3$, $e_2=7$, $C_1=147$, $C_2=279$.

Since $147^{(-1)} \bmod 391 = 133$, $M = 279 * (147^{-1})^2 \bmod 391 = 279 * 133^2 \bmod 391 = 29$.

You can verify that indeed $29^3 \bmod 391 = 147$, and $29^7 \bmod 391 = 279$.

Give **bonus** 4 points for checking and pointing out that $335^3 \bmod 391 = 334$, not 74, and $335^7 = 262$, not 279

	Quotient, Q(i)	Remainder, R(i)	S(i)	T(i)
		$R(i+1) = R(i-1) - Q(i+1)R(i)$	$S(i+1) = S(i-1) - Q(i+1)S(i)$	$T(i+1) = T(i-1) - Q(i+1)T(i)$
0		391	1	0
1		74	0	1
2	5	21	1	-5
3	3	11	-3	16
4	1	10	4	-21
5	1	1	-7	37

Marks
on Q9

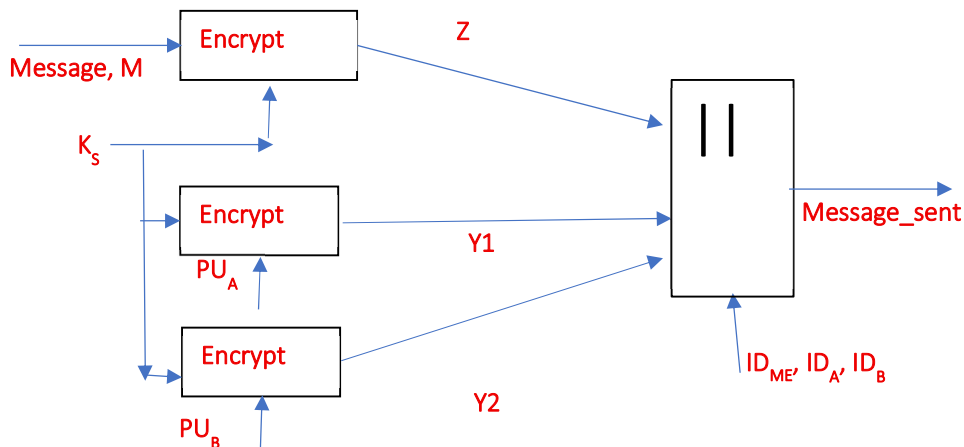
9. I have two PhD students, viz. A and B. Each has been assigned a public-private key pairs, viz. $\langle PU_A, PR_A \rangle$ and $\langle PU_B, PR_B \rangle$. They have securely shared their public keys with me. I can now encrypt and confidentially share messages with student A using his public key, PU_A , and similarly with student B using public key, PU_B . Often, I want to encrypt and copy the same message to them in one email. Invariably the messages are very long. Can I use symmetric key cryptosystem to encrypt the message using a symmetric key, and copy the same to both my PhD students. Of course, I must somehow ensure that they are able to obtain the symmetric key and thereby decrypt the message.

a. Elaborate on how this can be achieved.

- generate a new "session key", say K_s , and use that to (symmetric-key) encrypt message M , to obtain $E(K_s, M)$.
- Encrypt (public-key) K_s using A's public-key PU_A , to obtain $E(PU_A, K_s)$.
- Encrypt (public-key) K_s using B's public-key PU_B , to obtain $E(PU_B, K_s)$.
- Send to A and B encrypted message $E(K_s, M)$, together with $E(PU_A, K_s)$ and $E(PU_B, K_s)$, copies of session key K_s , suitably encrypted for A for B to recover the session-key.
- At A's end, for instance, A will decrypt the encrypted session key, $E(PU_A, K_s)$, using his private key PR_A , to obtain the session key K_s and use that to decrypt the message, M .

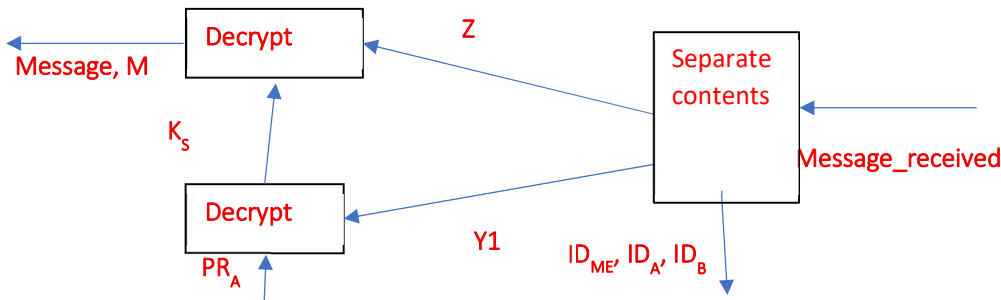
5 points for
the above

b. Use a block diagram to show the processing at my end.



3 points

c. Use a block diagram to show the processing at my student A's end.



2 points

d. Give the contents of the message that I send them. The format you should use is given below.

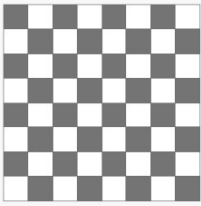
Message_sent = $\langle ID_A, ID_B, \dots, E(\dots, \text{Long_file}) \rangle$

Message_sent = $\langle ID_{ME}, ID_A, ID_B, E(PU_A, K_s), E(PU_B, K_s), E(K_s, M) \rangle$

2 points

10. I have a Black-and-White photo (with no grayscale) of an 8x8 chess board. The photo is 1024x1024 pixels where each pixel is encoded using 8 bits (see sample photo below, but ignore the lines on the edge of the board).

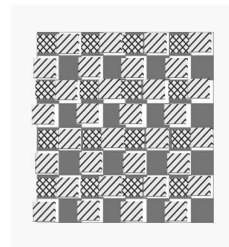
Marks
on Q10



I wish to encrypt this image and communicate the resulting ciphertext confidentially so that an intruder is NOT able to determine that this is a ciphertext corresponding to a chess board.

- A. Can I do this using a simple block cipher?
- YES/NO **NO, using for instance DES or AES block ciphers**
 - If this is to succeed, should there be any constraint on encryption algorithm? Please be specific.
 - It will NOT work if the standard block size is 64 or 128 bits. The block size should be very large.**
 - What does the ciphertext look like when this method is used?

3 points for
part A,
equally
distributed
between the
3 sub-parts

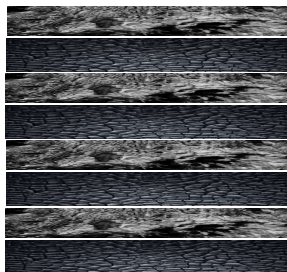


- B. Can I do this using a block cipher with block chaining?
- YES/NO **YES, even with DES/AES with block chaining**
 - If this is to succeed, should there be any constraint on encryption algorithm? Please be specific.
 - No particular constraint is necessary.**
 - What does the ciphertext look like when this method is used?



4 points for part B,
distributed
between the 3
sub-parts as 1, 1, 2

- C. Can I do this using a stream cipher?
- YES/NO **Yes, under some constraint on periodicity of sequence of random numbers.**
 - If this is to succeed, should there be any constraint on encryption algorithm? Please be specific.
 - None on the algorithm. But, YES, there is some constraint on periodicity of random numbers.**
 - What does the ciphertext look like when periodicity of PRSG stream is 1024? OR when periodicity of PRSG stream is $1024 * 1024$?



When periodicity is 1024



When periodicity is $1024 * 1024$

5 points for part C,
distributed between the 3
sub-parts as 1, 1, and 3
(this for the 2 pictures)